

BEOGRAD SURCIN, Serbia

WMO: 132720

Lat: **44.818N**Lon: **20.309E**Elev: **102**StdP: **100.10**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-7.8	-12.9	1.2	-8.9	-10.1	1.6	-6.6	11.4	2.9	10.2	2.8	2.9	110	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	34.8	21.4	32.9	21.4	31.1	20.7	23.1	31.1	22.3	30.0	21.5	28.8	3.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.8	15.7	26.3	19.9	14.8	25.4	19.1	14.0	24.3	68.8	31.2	65.8	30.1	63.1	28.8	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.5	7.3	-14.3	37.4	4.4	2.1	-17.4	38.9	-20.0	40.1	-22.5	41.3	-25.7	42.8		
(5)			-14.6	24.6	4.4	0.9	-17.8	25.3	-20.3	25.8	-22.8	26.3	-25.9	26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	1.1	3.3	7.6	12.9	17.6	21.1	23.0	23.0	18.2	13.1	7.9	2.2		
	DBStd	8.88	4.95	5.45	4.78	4.16	3.80	3.73	3.34	3.37	3.84	4.43	4.89	4.86		
	HDD10.0	959	278	194	103	20	1	0	0	0	0	21	98	244		
	HDD18.3	2571	535	422	333	167	61	16	4	4	50	168	314	499		
	CDD10.0	1919	2	6	29	108	236	334	402	403	244	117	34	3		
	CDD18.3	488	0	0	0	4	38	100	147	149	44	6	0	0		
	CDH23.3	5112	0	0	2	57	362	1032	1590	1597	413	58	1	0		
	CDH26.7	2052	0	0	0	8	84	401	701	726	127	6	0	0		
(14) Wind	WSAvg	3.1	3.2	3.5	3.7	3.5	3.0	2.8	2.7	2.5	2.8	3.0	3.3	3.1		
(15) Precipitation	PrecAvg	667	44	39	46	50	66	84	67	53	63	54	48	52		
	PrecMax	1011	85	89	118	148	244	175	235	128	187	115	134	140		
	PrecMin	352	8	1	6	3	17	18	11	3	10	0	3	5		
	PrecStd	159	21	21	32	29	43	46	55	35	36	33	28	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	18.9	22.8	28.0	31.1	34.8	37.0	37.8	33.1	27.2	22.2	16.2		
		MCWB	10.4	11.5	13.5	17.0	19.3	21.6	21.1	21.6	19.9	18.3	14.9	11.5		
	2%	DB	12.0	15.9	20.2	24.9	29.0	32.8	34.8	35.0	30.0	25.0	19.8	13.7		
		MCWB	8.7	10.2	11.9	15.6	18.7	21.9	21.5	21.5	18.9	17.0	14.0	9.8		
	5%	DB	10.1	13.2	18.0	22.6	27.1	31.0	32.8	32.9	27.8	22.8	17.1	11.1		
		MCWB	7.6	8.8	11.2	14.2	18.3	21.0	21.6	21.1	18.5	15.9	12.5	8.3		
	10%	DB	8.0	11.0	15.6	20.4	25.2	29.0	30.8	30.9	25.8	20.2	15.0	8.9		
		MCWB	6.2	7.6	9.9	13.1	17.4	20.3	20.9	20.4	17.7	15.0	11.5	7.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	12.0	14.2	17.7	20.8	24.1	23.9	24.5	21.7	19.6	15.8	12.1		
		MCDB	14.0	17.6	21.0	25.7	27.6	31.3	32.3	31.9	30.1	25.8	20.9	15.8		
	2%	WB	9.1	10.5	12.8	16.2	19.6	22.7	23.0	23.0	20.2	17.7	14.2	10.1		
		MCDB	11.7	15.1	18.7	23.6	26.9	30.1	30.9	31.5	27.1	22.7	18.8	13.0		
	5%	WB	7.7	9.2	11.5	15.0	18.7	21.8	22.3	22.0	19.2	16.5	13.0	8.6		
		MCDB	9.7	12.8	16.9	21.3	25.8	29.0	30.0	29.6	25.4	21.4	16.5	11.0		
	10%	WB	6.2	7.8	10.2	13.8	17.8	20.9	21.5	21.1	18.2	15.4	11.7	7.0		
		MCDB	7.7	10.6	14.9	19.0	24.2	27.7	28.6	28.6	24.2	19.8	14.8	8.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	7.9	9.7	10.5	10.8	10.9	11.5	11.8	10.6	10.0	8.0	6.0		
		MCDBR	9.7	11.9	13.6	14.1	13.3	13.8	14.7	15.1	14.0	13.3	11.5	9.5		
	5% WB	MCWBR	7.0	7.3	7.3	6.9	5.7	5.3	5.1	5.2	5.5	6.6	6.7	6.6		
		MCWBR	9.2	11.0	12.4	12.5	12.6	12.6	12.8	12.9	12.3	12.1	10.6	8.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.346	0.373	0.404	0.461	0.448	0.444	0.447	0.456	0.435	0.399	0.361	0.344		
		taud	2.334	2.246	2.210	2.061	2.161	2.212	2.200	2.175	2.201	2.296	2.357	2.351		
	Ebn at Noon	Ebn at Noon	749	794	820	801	826	830	821	798	779	756	725	710		
		Edh at Noon	81	105	124	157	146	140	140	138	124	99	79	73		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.24	2.06	3.29	4.53	5.52	6.12	6.21	5.51	3.86	2.65	1.61	1.04		
		RadStd	0.17	0.27	0.39	0.46	0.53	0.57	0.47	0.52	0.42	0.29	0.14	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.71	N/A	N/A	N/A	N/A	N/A	N/A	-162	+125	N/A
(47) Regional (50 neighbors)		+0.67	N/A	N/A	N/A	+0.36	N/A	-116	-178	+119	+63

Nomenclature: See separate page